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REVIEW

Progress in the application of machine learning in CT diagnosis of acute appendicitis

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Abstract

Acute appendicitis represents a prevalent condition within the spectrum of acute abdominal pathologies, exhibiting a diverse clinical presentation. Computed tomography (CT) imaging has emerged as a prospective diagnostic modality for the identification and differentiation of appendicitis. This review aims to synthesize current applications, progress, and challenges in integrating machine learning (ML) with CT for diagnosing acute appendicitis while exploring prospects. ML-driven advancements include automated detection, differential diagnosis, and severity stratification. For instance, deep learning models such as AppendiXNet achieved an AUC of 0.81 for appendicitis detection, while 3D convolutional neural networks (CNNs) demonstrated superior performance, with AUCs up to 0.95 and an accuracy of 91.5%. ML algorithms effectively differentiate appendicitis from similar conditions like diverticulitis, achieving AUCs between 0.951 and 0.972. They demonstrate remarkable proficiency in distinguishing between complex and straightforward cases through the innovative use of radiomics and hybrid models, achieving AUCs ranging from 0.80 to 0.96. Even with these advancements, challenges remain, such as the "black-box" nature of artificial intelligence, its integration into clinical workflows, and the significant resources required. Future directions emphasize interpretable models, multimodal data fusion, and cost-effective decision-support systems. By addressing these barriers, ML holds promise for refining diagnostic precision, optimizing treatment pathways, and reducing healthcare costs.

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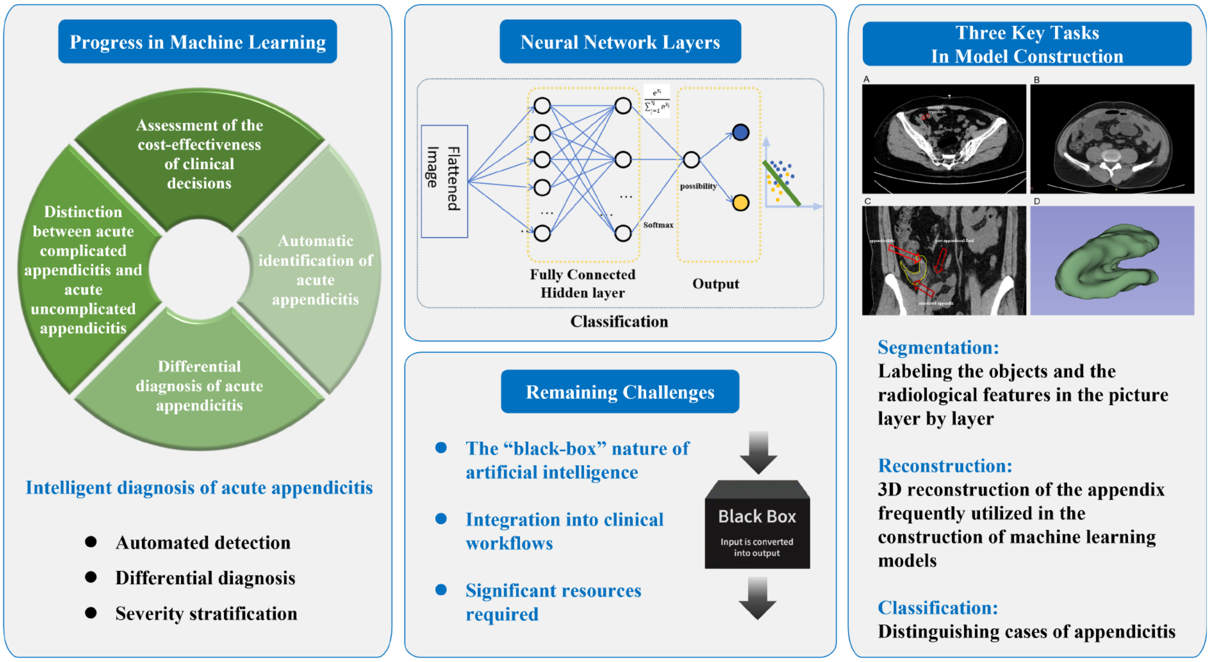
Graphical Abstract



Keywords Acute appendicitis · Machine learning · Computed tomography

Introduction

In the context of relentless advancements in medical imaging technology, Computed Tomography (CT) has become an indispensable diagnostic modality for acute appendicitis [1]. The evolution of CT imaging, marked by enhanced resolution, rapid acquisition protocols, and 3D reconstruction capabilities, has generated vast repositories of high-quality anatomical data. These advancements not only improved diagnostic precision but also laid the foundation for data-driven approaches, paving the way for artificial intelligence (AI) to address persisting challenges in imaging interpretation [1, 2]. Over the past few years, the evolution of artificial intelligence has spurred the extensive application and implementation of machine learning in medical imaging. Early AI applications in this field include Forsström et al.'s 1995 study, which used single-layer perceptrons and neuro-fuzzy logic to diagnose appendicitis, achieving an AUC of 0.6825 [3]. More recently, Park et al. (2020) integrated 3D convolutional neural networks (CNNs) with CT data, significantly improving diagnostic accuracy [4]. This article endeavors to provide an in-depth insight into the evolution of machine learning applications, specifically focusing on their role in enhancing the diagnostic accuracy of acute appendicitis through CT imaging.

Overview of acute appendicitis

Acute appendicitis is recognized as the leading cause of emergent abdominal surgical procedures throughout the world and presents as a chief complaint in emergency department consultations, representing 10–15% of total abdominal surgical cases. The worldwide incidence is about 7–8%. Males exhibit a marginally higher prevalence, with an estimated lifetime risk of 8.6%, in contrast to the 6.7% observed in females [5–10]. The condition is predominantly precipitated by bacterial invasion and obstruction within the appendiceal lumen. If not diagnosed and treated promptly, acute appendicitis can lead to complications such as purulent appendicitis and perforation. In severe cases, it may progress to sepsis, posing a threat to the patient's life [11–15]. Due to the variable location of the appendix and the diverse clinical presentations, acute appendicitis can be easily confused with other conditions that cause acute abdominal pain, which poses significant challenges for clinical diagnosis and treatment [16].

Acute appendicitis can be classified into acute uncomplicated appendicitis and acute complicated appendicitis [17]. Acute uncomplicated appendicitis, a reversible condition, is characterized by the absence of gangrene or necrosis within the appendix, with the majority of cases being amendable to antibiotic therapy. Conversely, acute complicated appendicitis is characterized by appendiceal gangrene, necrosis, perforation, and the presence of abscesses with purulent exudate accumulation within the abdominal and pelvic cavities [11, 18, 19]. The diagnostic and therapeutic strategies for these distinct forms of acute appendicitis diverge. In general, empirical broad-spectrum antibiotic therapy is recommendable for acute uncomplicated appendicitis, such as monotherapy with ertapenem, or a combination therapy with cephalosporins plus metronidazole intravenously [18, 20]. In contrast, acute suppurative or gangrenous appendicitis necessitates urgent surgical intervention for appendectomy to preempt perforation and potentially life-threatening complications. When the condition progresses to peri appendiceal abscesses, it is necessary for the performance of ultrasound-guided percutaneous drainage [12–15, 20].

CT in the diagnosis of acute appendicitis

CT imaging constitutes a pivotal diagnostic approach for acute appendicitis, with its expedited scanning capabilities reducing the examination duration for patients with abdominal pain, consequently minimizing their distress [21]. Kaur Mohi et al. [22]compared ultrasound and CT scans in diagnosing acute appendicitis, and found CT scans more accurate (90% vs. 84%) and specific (100% vs. 75%) than ultrasound. The utilization of multi detector row helical CT scanning in conjunction with three-dimensional (3D) reconstruction techniques facilitates a comprehensive assessment of the appendix and its surrounding anatomical structures from diverse perspectives, yielding imagery that closely mirrors the intraoperative findings, thereby enhancing the precision of clinical diagnostic and therapeutic protocols [23, 24].Research has identified that the CT grading system is also predictive of the pathological outcomes following appendectomy for acute appendicitis. This finding is instrumental in discerning the pathological subtypes of acute appendicitis and in informing the selection of appropriate treatment modalities [10, 25].

CT can also be instrumental in assessing the severity of acute appendicitis, thereby facilitating the formulation of precise diagnostic and therapeutic strategies by clinicians. Extensive research on the CT imaging characteristics of acute appendicitis has been conducted. The CT manifestations of acute uncomplicated appendicitis encompass dilation of the appendix, enhancement of the appendiceal wall thickness, and inflammatory stranding of the peri-appendiceal fat tissues, among others. Conversely, the CT features of acute complicated appendicitis include the presence of extraluminal appendicoliths, formation of abscesses, defects in the appendiceal wall, extraluminal gas, intestinal obstruction, accumulation of fluid in the peri-appendiceal or free abdominal cavity, and severe inflammation or cellulitis surrounding the appendix [18]. A study by Song et al. has demonstrated a correlation between the infiltration of peri-appendiceal fat and the severity of acute appendicitis [26]. Furthermore, a single-center study by Horrow et al. has identified that the presence of appendiceal wall defects, severe inflammation or cellulitis, abdominal cavity fluid, extraluminal gas, or extraluminal appendicoliths exhibit a high degree of specificity for the diagnosis of appendiceal perforation, with severe inflammation or cellulitis showing a specificity of 95%, and the rest exhibiting a specificity of 100% [27, 28].

Progress in machine learning in the diagnosis of acute appendicitis on CT

Intelligent diagnosis of acute appendicitis

Machine learning, particularly deep learning technology, has demonstrated significant prowess in image recognition. This technology has the potential to autonomously identify lesion within the appendiceal region on CT scans. Consequently, it executes the role of a valuable adjunct to radiologists, facilitating the expedited and precise determination of diagnostic outcomes. The radiological features of appendicitis, such as appendicoliths and peri-appendiceal fluid, are essential for ML model training and validation. The application of such algorithms is particularly pertinent to the diagnosis of acute appendicitis, the differentiation from other pathologies, and the stratification of disease complexity (Fig. 1).

Automatic identification of acute appendicitis

Machine learning models offer a timely and precise method for identifying appendicitis in CT scans. These models can recognize subtle changes that may go unnoticed by human radiologists. This technological advancement significantly enhances diagnostic accuracy and contributes to improved patient care outcomes [4, 29, 30]. Moreover, in clinical practice, radiologists can integrate ML tools into their routine work to assist in diagnosing appendicitis [31, 32]. For instance, Rajpurkar et al. [29]developed 3D ResNet model with 438 CT scans, achieving an AUC of 0.810 for acute appendicitis diagnosis. Another study by Park et al. [4]involved 667 subjects with acute appendicitis, using a 3D CNN algorithm for classification and conducted external validation in real world, achieving high classification efficiency. Juho et al. [30] highlighted the potential of integrating Automated Machine Learning (AutoML) and Automated Feature Engineering (AutoFE) into clinical diagnosis, especially equivocal appendicitis, achieving an

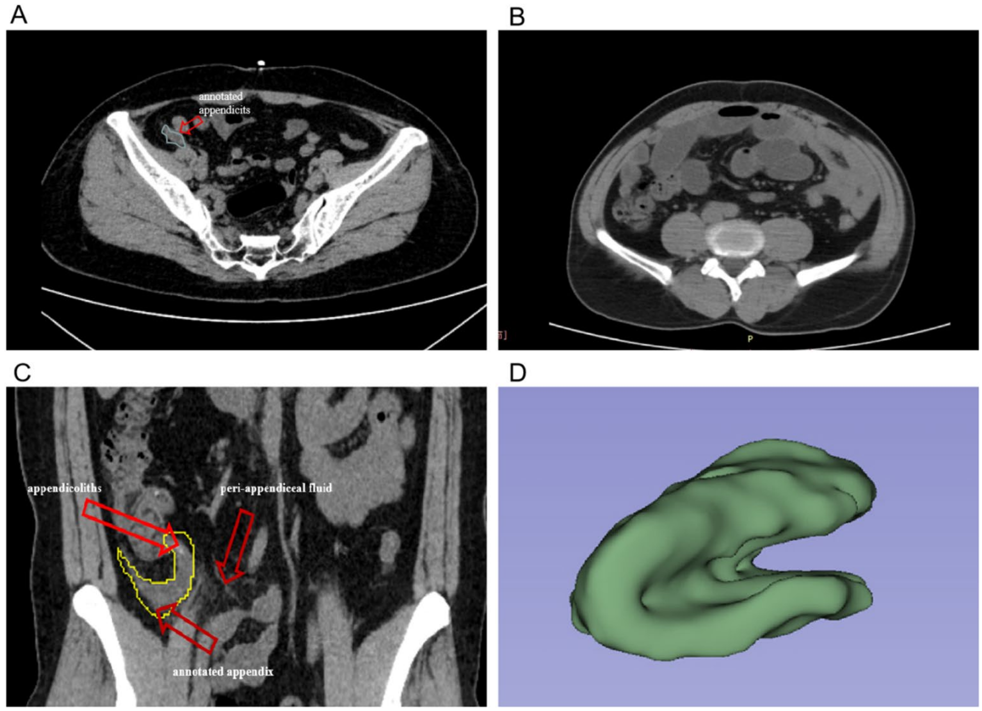


Fig. 1 **A**and **B** depict cross-sectional CT images of a patient with appendicitis and a normal individual, respectively. By comparison, the appendix in Fig. **A** is visibly larger than that in **B**. Moreover, peri-appendiceal fluid is present in **A**, whereas it is absent in **B**. **C** illustrates the radiological features associated with appendicitis, including appendicoliths and peri-appendiceal fluid. It also displays the corresponding annotations for appendiceal segmentation in CT imaging, which are commonly employed in machine learning applications. **D** presents a 3D reconstruction of the appendix in a case of appendicitis, a technique frequently utilized in the construction of machine learning models

AUC of 0.886 and an accuracy of 0.915. Dogan et al. [33] also developed a deep learning approach using hybrid CNN models and ensemble learning, achieving 0.833 and 0.96 accuracy in diagnosing acute appendicitis from CT images in challenging and clear cases, respectively. In comparison, Park et al. [4] achieved highest AUC(AUC = 0.95) among these studies, which may contribute to following reasons. Park et al. conducted a multicenter investigation, whereas other studies were primarily single-center, which may have resulted in insufficient sample representativeness and a lower AUC. Most of the CT images utilized in the study of Park et al. were enhanced portal phase CT images (629/667) and data were relatively balanced. Research indicates that enhanced CT is more effective in diagnosing appendicitis, which may contributes to the increased diagnostic efficacy of ML [34–37]. Hariri et al. [38] achieves 99% accuracy highest among these studies, which may contribute to its a lightweight dual-path CNN. In a word, ML models could provide crucial secondary diagnostic insights for identifying uncertain cases when CT scans yield ambiguous results, reduce misdiagnoses, minimize unnecessary surgeries, and improve patient outcomes, enhancing diagnostic reliability and efficiency in healthcare.

Differential diagnosis of acute appendicitis

Acute appendicitis exhibits clinical manifestations that are often indistinguishable from other acute abdominal pathologies, such as diverticulitis. For instance, both conditions frequently present with pain localized to the lower right quadrant of the abdomen. This similarity complicates the differential diagnosis based on clinical grounds alone [39]. Moreover, the potential for misdiagnosis exists even with radiologists' interpretation of CT scans. Park et al. [40] used CNN to classify acute appendicitis, acute diverticulitis, and normal appendix, achieving 89.66% sensitivity, 93.62% accuracy, and 95.47% specificity for normal appendix identification, and 0.952 and 0.972 AUC values for appendicitis and diverticulitis, respectively, demonstrating AI's potential to improve diagnostic accuracy. Lee et al. [41] utilized a deep learning approach and RGB (Red, Green, Blue) composite images, surpassing the performance of traditional single-channel image-based classification (AUC = 0.967 vs. 0.959) in appendicitis and diverticulitis classification, comparable to that in Park et al. Prabhudesai et al. [42] employed artificial neural networks to construct a diagnostic model for appendicitis in patients with acute right iliac fossa pain, with sensitivity, specificity, positive predictive value, and negative predictive value of 100%, 97.2%, 96.0%, and 100%, respectively. The ability of the model to accurately exclude patients without appendicitis was significantly superior to that of experienced clinicians and the Alvarado scoring system.

The distinction between acute complicated appendicitis and acute uncomplicated appendicitis

The accurate grading of acute appendicitis is crucial for clinicians to formulate appropriate treatment protocols, which can, in turn, improve therapeutic efficacy. However, differentiating between acute complicated appendicitis and acute uncomplicated appendicitis preoperatively remains a formidable challenge. Many researchers endeavor to address this issue.

Machine learning models typically incorporate CT features including intraluminal gas, intraluminal fluid, appendicoliths, peri-appendiceal fluid collection, stranding, cecal wall thickening, and enlarged ileocecal lymph nodes. Researchers often utilized convolutional layers to detect spatial patterns in CT slices and quantified through texture analysis. Study have demonstrated that machine learning models incorporating these features exhibit superior classification performance [43–48]. Byun et al. [43] constructed a decision tree model to distinguish appendicitis with and without complications with appendiceal abscess, peri-appendiceal inflammatory mass, and free air as the principal features, exhibiting an AUC of 0.7 in the test cohort. Kim et al. [44] included more imaging features such as appendiceal wall enhancement defects, abscess formation, peri-appendiceal fat stranding appendiceal diameter, and the presence of extraluminal air and achieved an higher AUC of 0.81.

To distinguish complicated from uncomplicated appendicitis, aiming to reduce adverse outcomes, researchers have used various methods, including radiomics, to improve machine learning accuracy. Radiomics involves extracting high-throughput features from CT images to enhance predictive accuracy [49]. Zhao et al. [45] applied radiomics to develop a model to differentiate between complicated and uncomplicated appendicitis, demonstrating superior diagnostic performance compared to models based solely on CT features (AUC = 0.804 vs. 0.669).

Radiologists frequently seek decision-support tools that provide clear and transparent rationales for their recommendations, ensuring that AI outputs are interpretable is essential for fostering trust and facilitating integration into clinical practice [50]. Wei et al [51]. performed well (accuracy = 95.56%) in differentiating between complicated and uncomplicated appendicitis with the Gradient Boosting Machine, including predictors such as total bilirubin, abdominal pain duration, and temperature. The model's predictions became further transparent using Shapley Additive Explanations technology, enhancing transparency. Males et al. [52] developed an interpretable ML model involving 551 patients to reduce unnecessary appendectomies in children with suspected acute appendicitis. The model achieved a sensitivity of 99.7% and a specificity of 17%, outperforming traditional risk assessment tools like that in Wei et al. [51], indicating the model could lower unwarranted surgeries while maintaining high diagnostic accuracy. These models can be used during night shifts in the emergency department, reducing the incidence of complications and mortality.

Assessment of the cost-effectiveness of clinical decisions

Machine learning empowers the selection of personalized diagnostic and treatment plans by expertly analyzing clinical indicators, CT images, lab results, and follow-up data, ensuring that each patient's unique profile is meticulously considered. This methodology can diminish expenditure on diagnostic procedures and associated healthcare costs, thereby augmenting the economic efficacy of clinical decision-making. Gregory et al. [53] evaluated a comprehensive strategy combining clinical decision rules (CDR) with stratified imaging protocols for pediatric appendicitis diagnosis using a Markov model. The integrated approach was the most cost-effective, reducing CT scan utilization and associated costs in simulated cases of 100,000 10-year-olds in US emergency departments. Furthermore, Viradia et al. [54] identified CT imaging characteristics as independent predictors of hospital resource utilization in acute appendicitis patients. Higher Charlson Comorbidity Index, increased appendiceal wall thickness, retroperitoneal fluid, and fecalith were associated with repeated CT scans and hospital revisits. Larger appendiceal diameter and free intra-abdominal air correlated with higher hospitalization costs. However, the predictive model had limitations for cost estimates exceeding $45,000.

Multidimensional data-driven appendicitis diagnostic model

By analyzing multidimensional data such as demographic information, clinical characteristics, and laboratory test results, it is possible to discern the characteristic patterns of appendicitis and construct a more comprehensive diagnostic model. Employing machine learning methods to analyze the relevant data can further enhance the model's accuracy. In 1986, Alvarado [55] proposed a scoring system for the diagnosis of acute appendicitis, which included eight predictive factors: localized tenderness in the lower right abdomen, leukocytosis, migratory pain, left-sided metastatic pain, fever, nausea and vomiting, anorexia, and rebound tenderness. Ting et al. [56] optimized the Alvarado scoring system using a decision tree model and found that of the new model could reach, respectively, significantly outperforming the original scoring system (sensitivity = 0.945;specificity = 0.805). Kang et al. [57] used a decision tree algorithm incorporating the severity of rebound tenderness, symptom duration, white blood cell count, neutrophil count, and C-reactive protein to model, achieving a higher AUC compared with the Alvarado score (0.695 vs. 0.85.) Eickhoff et al. [58] constructed a machine learning model that includes various demographic, clinical characteristics, and laboratory indicators, which can predict postoperative complications at an individual level, such as intensive care treatment (accuracy = 0.68) and prolonged hospital stay (accuracy = 0.76). Bunn [59] utilized real-world public data in National Surgery Quality Improvement Program for model construction and identified preoperative risk factors associated with postoperative sepsis in appendectomy such as heart failure, blood transfusion, and acute renal failure through ensemble learning.

Recently, some scholars have attempted to combine multiple data sources such as text, images, and numbers to construct a multimodal AI to enhance the model's intelligence and performance. This study used machine learning to predict the severity of acute appendicitis. Schipper et al. [60]used logistic regression and K-Nearest Neighbors to predict acute appendicitis severity with 96% accuracy using age, C-reactive protein, and periappendiceal fluid collection. Liang et al. [46] developed a diagnostic model based on selected clinical characteristics, CT visual features, deep learning features, and radiomics features to distinguish between acute uncomplicated and complicated appendicitis, significantly better than other models and radiologists' visual diagnosis (AUC = 0.799 vs. 0.679). Li et al. [61] used a decision tree model to establish a scoring system for complicated and uncomplicated appendicitis during pregnancy based on clinical and imaging features, with an AUC of 0.78.

Summary and prospects

In conclusion, the application of machine learning to CT-based diagnostic models for appendicitis has achieved considerable advancements, contributing to the enhancement of diagnostic accuracy and efficiency, as well as the reduction of medical risks and associated costs. Nevertheless, several challenges remain that warrant further investigation.

One of the primary obstacles is the integration of ML models into existing appendicitis diagnosis radiology workflows. Radiologists typically use picture archiving and communication systems for diagnosis, while adding AI as a third component may complicate this process. Thus, it is essential adopt AI-assisted imaging systems that streamline diagnostic workflows and enhance the accuracy of appendicitis diagnosis and classification. Besides, establishing effective feedback loops between radiologists and AI systems is essential for continuous AI improvement [62]. However, this requires a structured approach to data collection and analysis that may not currently exist in many practices [62, 63]. Integration also requires systems with enhanced generalizability. Current model varied significantly across different institutions. For example, the difference of AUC can be up to 0.1 in different centers, indicating notable differences in the model's predictive accuracy across institutions [46]. This endeavor requires the advancement of inter-institutional data-sharing initiatives, the creation of multicenter databases, the optimization of image feature extraction methodologies, and the addressing of ethical considerations about the clinical deployment of AI models.

Besides, the accuracy and interpretability of AI in the context of acute appendicitis diagnosis and treatment remain areas that necessitate further enhancement. Despite existing efforts in this domain, including the application of SHapley Additive exPlanations algorithms, Gradient-weighted Class Activation Mapping and other interpretable methods [29, 51, 52], the inherent "black-box" nature of numerous AI algorithms still raises concerns in medical diagnosis, as physicians need to comprehend how and why a model reaches a specific conclusion [62, 64–67]. In future, AI models should provide radiologists with clear conclusion that can be weighed against their clinical judgment. For the precision of the AI, Future studies should include both appendiceal and extra-appendiceal regions in labeling to improve diagnostic accuracy. The current focus on the appendiceal region alone may miss complications like peritonitis, thereby reducing model discriminative power [45, 46].

Additionally, the implementation of ML technologies often requires significant investment in high-performance computing resources infrastructure, and labor costs. Data management, including acquisition, cleaning, and processing, is resource-intensive. Model training and validation require substantial computational resources and rigorous evaluation. Image segmentation is also labor-intensive [68, 69]. Thus, unsupervised, self-learning algorithms for image segmentation could represent a significant breakthrough in machine learning-assisted appendicitis diagnosis and treatment. By reducing the need for manual imaging analysis, radiologists can devote more time to patient care and follow-up evaluations, especially beneficial in high-volume clinics, where large datasets are collected. In conclusion, despite substantial initial investments in AI technologies, these costs could be offset over time through accurate disease detection with increased diagnostic efficiency.

Balancing opinions between ML models and radiologists is essential for optimizing diagnostic accuracy and patient outcomes. Disagreement prediction models can be employed to identify cases where AI and radiologists are likely to diverge, flagging these for further review and mitigating risks associated with automation bias [70]. Continuous feedback loops between radiologists and AI systems facilitate iterative improvements, like updating the data set dynamically, to refine algorithms based on clinical feedback [70–72]. Training programs focused on AI integration can equip radiologists with the skills to interpret and collaborate effectively with ML models. Close collaboration with IT professionals can help radiologist to address technical challenges and streamline data flow between systems [73]. Finally, ethical challenges also existed, especially in patient data privacy during ML model training, using data beyond initial consent, privacy risks like data memorization and membership inference attacks, and real world application risks in healthcare, and legal and financial sectors [74, 75]. Thus, ethical guidelines must be developed to emphasize transparency, patient safety, and human oversight, fostering a collaborative rather than competitive environment [76, 77].

Currently, most applications of machine learning in the clinical management of appendicitis are limited to diagnosis and differential diagnosis. In the future, research could extend the use of these models to a wider range of aspects. For example, they can be utilized to predict treatment outcomes such as relapse rates and the possibility of surgical complications such as surgical site infections, bowel obstructions or sepsis. Beyond treatment outcome prediction, these models could be optimized to guide personalized surgical method and timing decisions, evaluate intervention risks for specific patient subgroups, and forecast recovery state based on preoperative condition. Realizing these applications will require establishing standardized protocols for capturing multi-center clinical data including perioperative variables, longitudinal recovery data, and patient-reported functional status. Moreover, advanced algorithms capable of processing temporal data sequences, interpreting multimodal imaging features, and identifying cross-modal clinical correlations will be essential. Such multidimensional enhancements could establish an evidence-driven framework for personalizing surgical approaches, optimizing postoperative surveillance protocols, and tailoring rehabilitation plans, ultimately enhancing the clinical management of appendicitis.

Conclusion

The application of machine learning in CT diagnosis of acute appendicitis has demonstrated strong capabilities in identifying appendicitis, differentiating it from other conditions, and distinguishing between complicated and uncomplicated cases. These advancements help reduce misdiagnoses and unnecessary surgeries, ultimately improving patient outcomes.

However, challenges remain in integrating ML models into clinical workflows, ensuring model interpretability, and enhancing generalizability across institutions. Future research should focus on expanding the use of ML to predict treatment outcomes and guide surgical decisions, while addressing economic and technical barriers to implementation. Overall, while substantial progress has been made, continued efforts are needed to fully realize the potential of ML in the clinical management of acute appendicitis.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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